

String

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Introduction:

- String is a collection of characters that is treated as a single data item.
- In other words, string is one dimensional array of characters terminated by null('\0') characters.

Eg. `char city[]={ 'P', 'O', 'K', 'h', 'A', 'R', 'A', '\0' };`

- In string, the last character is always null.

Declaration of string variables:

char string_Name [size];

- The size should be equal to maximum number of characters in the string plus one.
- Plus one for null character.
- The only thing special about this array is that an extra character, the null character tells us where the string's value ends.

Initialization:

➤String can be initialized as:

```
char city[8]={ 'p', 'o', 'k', 'h', 'a', 'r', 'a', '\0'};
```

```
char city[8]="Pokhara";
```

➤The elements of character array are stored in contiguous memory locations.

```
char str[8];
```

```
str="Hello"; //Illegal
```

```
char str[6]="Hello";
```

```
char str2[6];
```

```
str2=str; //illegal
```

Array name cannot be used as left operand of assignment operators.

Reading string

➤ scanf() can be used with %s to read string.

eg.

```
char address[30];  
scanf("%s", address);
```

Here, scanf () function cannot read a string after white space. We have already done this example.

Here is an example

➤ C support format specification known as edit set conversion code %[^\n] that can be used to read a line of words containing white spaces.

See how it works. We have also done this already.

String input and output function

➤ `getchar( )`

➤ `gets( )`

➤ `scanf( )`

➤ `putchar( )`

➤ `puts( )`

➤ `printf( )`

[Example 1](#)

[Example 2](#)

String library functions:

- `strlen(str)`: Returns the number of characters in the string, do not include null character.
- `strlwr( )`: convert string to lower case.
- `strupr( )`: convert string to upper case.
- `strcat( )`: appends one string at the end of other.
- `strcpy( )`: copies a string to another.
- `strncpy( )`: copies first n character of one string to another.
- `strcmp( )`- compares two string.
- `strncmp( )`
- `stricmp( )`/ `strcmpi( )`; compare two string regardless of case.
- `strnicmp( )`:
- `strrev( )`: reverse the string.

Examples:

1. WAP to reverse a given string without using string reverse functions.

[program](#)

2. WAP to test whether the string is palindrome or not.

[program](#)

3. WAP to count number of vowels in given text.

[program](#)

4.WAP to generate:

N
NE
NEP
NEPA
NEPAL

[program](#)

